Faculty of Engineering & Technology

Department of Electrical & Computer Engineering

ENCS5344

Spoken Language Processing

Assignment 1

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Introduction

The goal of this project is to analyze the acoustic properties of five steady-state monophthong vowels (/i, e, a, o, u/) produced by three female speakers, and to compare acoustic measurements obtained through manual analysis in Praat with automatic analysis using Python's Parselmouth library. The project consists of three parts:

- **Part A:** Collection and acoustic analysis of vowels using Praat
- **Part B:** Extraction and interpretation of pitch (F_0) using Praat and Python
- **Part C:** Source–filter vowel synthesis and comparison between natural and synthetic vowels

The results provide insights into the acoustic structure of vowels, pitch patterns, and the relationship between natural and synthesized speech.

Part A: Data Collection & Acoustic Analysis

Three female speakers produced five vowels /i, e, a, o, u/ in hVd context, resulting in the words: heed, hayed, hard, hoed, who'd. Each vowel was repeated 10 times in a quiet room using phone microphones, and exported into 16 kHz, 16-bit WAV format.

Acoustic analysis was performed manually in Praat. For each token, the vowel portion was identified as the periodic, high-amplitude region between the /h/ and /d/. F1 and F2 were measured at the midpoint using: Formants → Show Formants → Get first formant / Get second formant.

Duration was measured using: Query → Get selection length. Spectrograms with formants displayed were saved for each vowel.

For each speaker and vowel, the mean F1, F2, and duration were calculated from a 5 repetitions.

Table 1: Raw F1, F2, and Duration Measurements for Speaker 1,2, and 3 Across Five Vowels

Speaker	Vowel	F1 (Hz)	F2 (Hz)	Duration (s)
Speaker 1	/a/	422.03894462566217 Hz	921.57521674099 Hz	0.132938
Speaker 1	/a/	384.7151248223316 Hz	936.2929520472775 Hz	0.109397
Speaker 1	/a/	467.35408949618807 Hz	867.6750007828156 Hz	0.114936
Speaker 1	/a/	408.3922419725302 Hz	919.9155316568476 Hz	0.103858
Speaker 1	/a/	468.2463627834421 Hz	887.8060479144301 Hz	0.108705
Speaker 1	/i/	417.0231391076271 Hz	2672.882206922092 Hz	0.171625
Speaker 1	/i/	384.79572907815435 Hz	2603.7334475933826 Hz	0.137034
Speaker 1	/i/	379.6692632739627 Hz	2798.2175936418994 Hz	0.130382
Speaker 1	/i/	398.29643778391454 Hz	2626.974419169016 Hz	0.149008
Speaker 1	/i/	389.79434272002453 Hz	2575.529972736409 Hz	0.143021
Speaker 1	/e/	500.8631461490959 Hz	2282.4608498558036 Hz	0.109375
Speaker 1	/e/	519.7484514759083 Hz	1941.9778296635461 Hz	0.110216

Speaker 1	/e/	489.1658312344116 Hz	2182.4608498558036 Hz	0.127884
Speaker 1	/e/	468.9141520451079 Hz	2437.652869474984 Hz	0.115685
Speaker 1	/e/	447.8816232844111 Hz	2487.5527495417427 Hz	0.128305
Speaker 1	/o/	447.8559677488725 Hz	879.9893890856775 Hz	0.120098
Speaker 1	/o/	445.8454246370693 Hz	865.9868459783828 Hz	0.110558
Speaker 1	/o/	448.41140787416447 Hz	820.4210752414865 Hz	0.140021
Speaker 1	/o/	427.2221142845603 Hz	843.2144237227833 Hz	0.092259
Speaker 1	/o/	441.56061413561343 Hz	822.9979144137342 Hz	0.111320
Speaker 1	/u/	375.331277772023 Hz	798.8705594099191 Hz	0.108756
Speaker 1	/u/	351.1752340185304 Hz	664.772201484347 Hz	0.129321
Speaker 1	/u/	369.01342935201615 Hz	541.6245693729853 Hz	0.102033
Speaker 1	/u/	364.0116569886284 Hz	691.233661729712 Hz	0.143162
Speaker 1	/u/	395.495802446895 Hz	844.5179224421024 Hz	0.128925
Speaker 2	/a/	580.6083291001066 Hz	1104.9228223257933 Hz	0.091265
Speaker 2	/a/	585.8583173729352 Hz	1089.800841721168 Hz	0.100172
Speaker 2	/a/	627.7614781837506 Hz	1094.514133697021 Hz	0.123129
Speaker 2	/a/	408.3922419725302 Hz	597.3857351016742 Hz	0.167789
Speaker 2	/a/	646.5419670101816 Hz	1101.382732706218 Hz	0.140659
Speaker 2	/i/	423.801854337113 Hz	2877.6336152956787 Hz	0.203437
Speaker 2	/i/	421.5276654200184 Hz	2767.7879405662725 Hz	0.189299
Speaker 2	/i/	420.4379746746343 Hz	2760.336060723517 Hz	0.198725
Speaker 2	/i/	410.3791621802593 Hz	2736.3710621564096 Hz	0.180659
Speaker 2	/i/	393.204366451449 Hz	2795.222207273984 Hz	0.214071
Speaker 2	/e/	529.8747793279796 Hz	2477.892085220126 Hz	0.157752
Speaker 2	/e/	570.981078733048 Hz	2415.283211074988 Hz	0.164325
Speaker 2	/e/	578.4864228178122 Hz	2492.4245388128625 Hz	0.190617
Speaker 2	/e/	563.3745678809279 Hz	2509.4708680749127 Hz	0.184865
Speaker 2	/e/	550.508652755987 Hz	2519.9412680952214 Hz	0.178292
Speaker 2	/o/	459.90862671251875 Hz	956.3802119959556 Hz	0.214401
Speaker 2	/o/	481.62110487529026 Hz	935.3975297819807 Hz	0.224840
Speaker 2	/o/	477.52774630634315 Hz	877.8810200811985 Hz	0.232870
Speaker 2	/o/	522.8832916597723 Hz	948.9917148547288 Hz	0.240097
Speaker 2	/o/	532.8604181272593 Hz	994.1153750993337 Hz	0.244915
Speaker 2	/u/	387.87808648013157 Hz	891.2586832500062 Hz	0.195240
Speaker 2	/u/	331.7485994617894 Hz	921.2690219976657 Hz	0.176299
Speaker 2	/u/	417.0438753536001 Hz	879.7970952678018 Hz	0.139145
Speaker 2	/u/	417.6251291793252 Hz	825.288023793506 Hz	0.182855
Speaker 2	/u/	411.4390798255819 Hz	909.2755011455479 Hz	0.177756
Speaker 3	/a/	800.5763346578889 Hz	1196.6893066091873 Hz	0.081689
Speaker 3	/a/	749.5183348116615 Hz	1180.5509849292703 Hz	0.097399
Speaker 3	/a/	620.9874525265649 Hz	1107.009650495565 Hz	0.120963
Speaker 3	/a/	643.3229443255229 Hz	1247.0258676819863 Hz	0.172019
Speaker 3	/a/	673.7385528734621 Hz	1240.8229718765465 Hz	0.167306
Speaker 3	/i/	381.08526340113076 Hz	2629.6264760326703 Hz	0.124136
Speaker 3	/i/	390.8042503394446 Hz	2609.769350891047 Hz	0.138487
Speaker 3	/i/	382.8245795110982 Hz	2434.5351538155496 Hz	0.152838
Speaker 3	/i/	411.0707041075389 Hz	2816.776104799688 Hz	0.147098
Speaker 3	/i/	418.5170768567364 Hz	2627.877772496404 Hz	0.171495
Speaker 3	/e/	646.2126397376081 Hz	2567.2779596871005 Hz	0.109989

Speaker 3	/e/	481.72348852246006 Hz	2574.377150742051 Hz	0.174337
Speaker 3	/e/	552.6593152529285 Hz	2499.6570934491615 Hz	0.166855
Speaker 3	/e/	545.6084356767897 Hz	2584.0507254375484 Hz	0.159372
Speaker 3	/e/	509.3711060007388 Hz	2531.1360142751714 Hz	0.170596
Speaker 3	/o/	545.9635152670027 Hz	1052.2097094361384 Hz	0.145079
Speaker 3	/o/	505.4622495251905 Hz	980.1833855640061 Hz	0.173939
Speaker 3	/o/	524.1963060983795 Hz	840.8049943434232 Hz	0.200459
Speaker 3	/o/	425.8445919779938 Hz	920.9125132608536 Hz	0.169259
Speaker 3	/o/	537.9630345844389 Hz	873.999423358242 Hz	0.199958
Speaker 3	/u/	316.5859566908968 Hz	891.7969541258835 Hz	0.101554
Speaker 3	/u/	443.6117380231363 Hz	720.5725494811688 Hz	0.092391
Speaker 3	/u/	404.34590105036864 Hz	1061.304413010636 Hz	0.143552
Speaker 3	/u/	429.5431357793069 Hz	1091.2564582193945 Hz	0.125150
Speaker 3	/u/	472.51760681386673 Hz	903.161213677955 Hz	0.097458

Table 2: Summary of Mean F1, Mean F2, and Mean Duration for All Speakers and Vowels

Speaker	Vowel	Mean F1 (Hz)	Mean F2 (Hz)	Mean Duration (ms)
Speaker 1	/a/	430.15 Hz	906.65 Hz	113.97 ms
Speaker 1	/i/	393.92 Hz	2655.47 Hz	146.21 ms
Speaker 1	/e/	485.31 Hz	2266.42 Hz	118.29 ms
Speaker 1	/o/	442.18 Hz	846.52 Hz	114.8512 ms
Speaker 1	/u/	371.01 Hz	708.20 Hz	122.44 ms
Speaker 2	/a/	569.8325 Hz	997.6013 Hz	124.60 ms
Speaker 2	/i/	413.87 Hz	2787.47 Hz	197.24 ms
Speaker 2	/e/	558.65 Hz	2483.00 Hz	175.17 ms
Speaker 2	/o/	494.96 Hz	942.55 Hz	231.42 ms
Speaker 2	/u/	393.15 Hz	885.38 Hz	174.26 ms
Speaker 3	/a/	697.63 Hz	1194.42 Hz	127.88 ms
Speaker 3	/i/	396.86 Hz	2623.72 Hz	146.81 ms
Speaker 3	/e/	558.65 Hz	2483.00 Hz	175.17 ms
Speaker 3	/o/	507.89 Hz	913.66* Hz	177.74 ms
Speaker 3	/u/	413.32 Hz	933.62 Hz	112.02 ms

The vowel space plots for Speakers 1–3 were generated using the **mean F1 and mean F2 values** from the summary table, allowing a clear visualization of each speaker's average vowel articulation patterns.

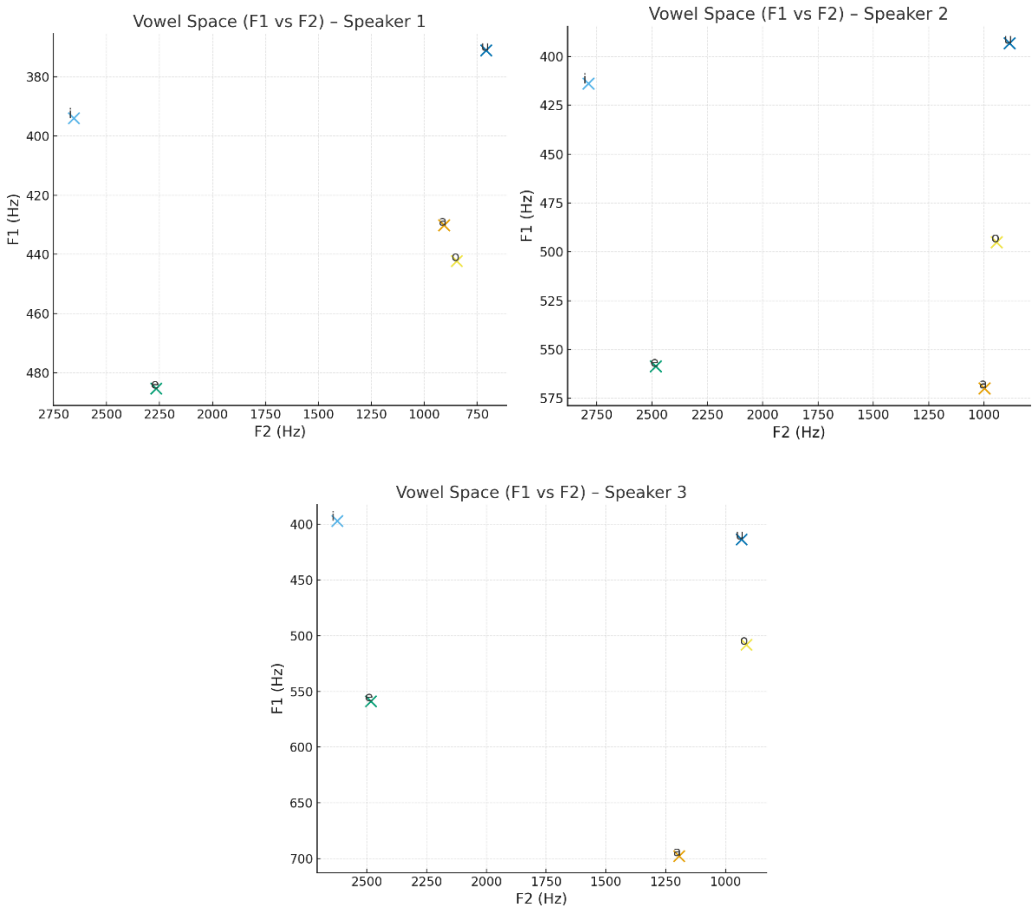

Figure 1: Vowel Space (F1 vs F2) for Speaker 1,2, and 3 Based on Mean Formant Values.

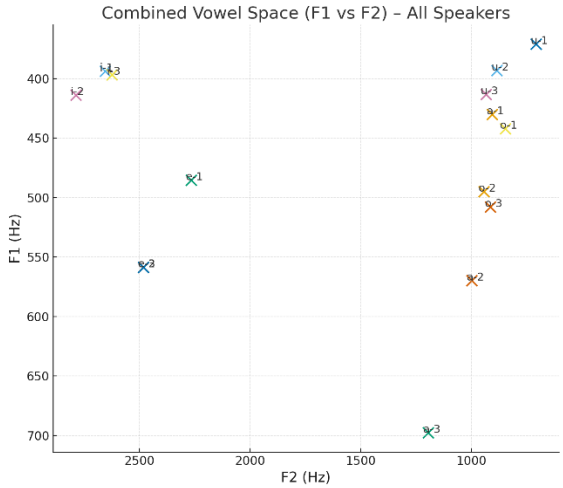

Figure 2: Combined Vowel Space Plot for All Speakers.

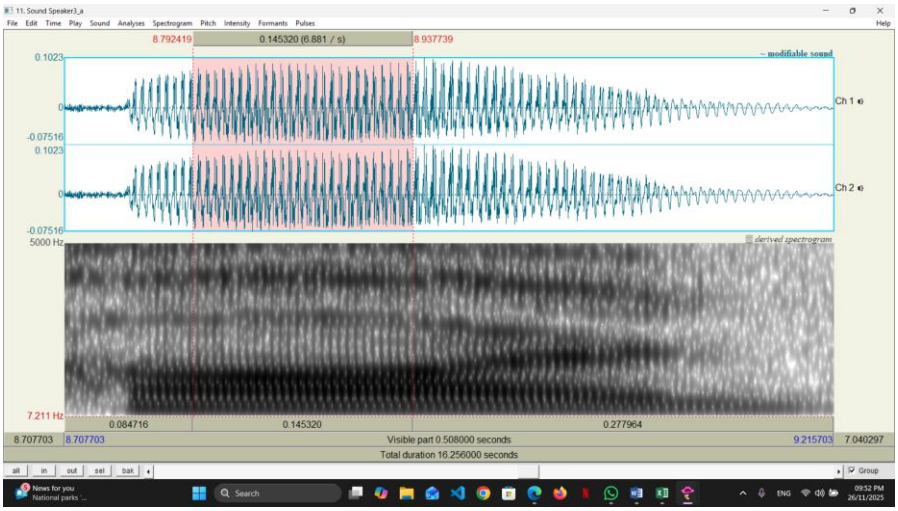

Figure 3: waveform and spectrum of vowel a

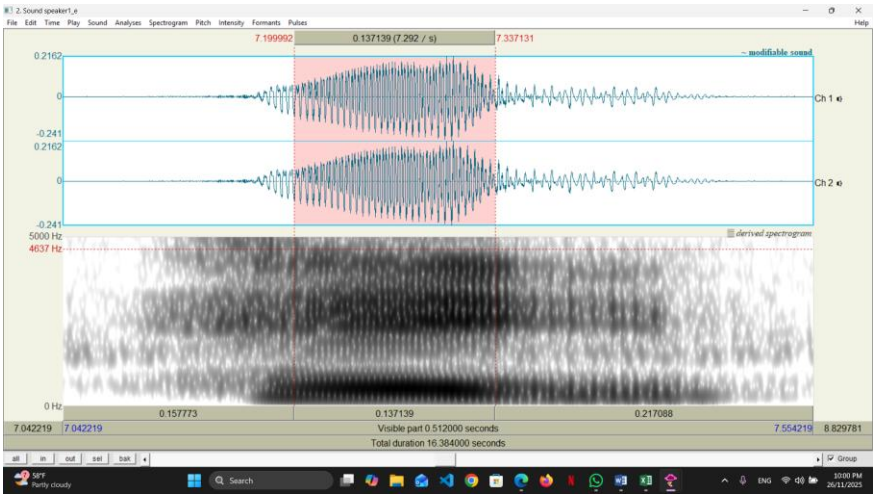

Figure 4: waveform and spectrum of vowel e

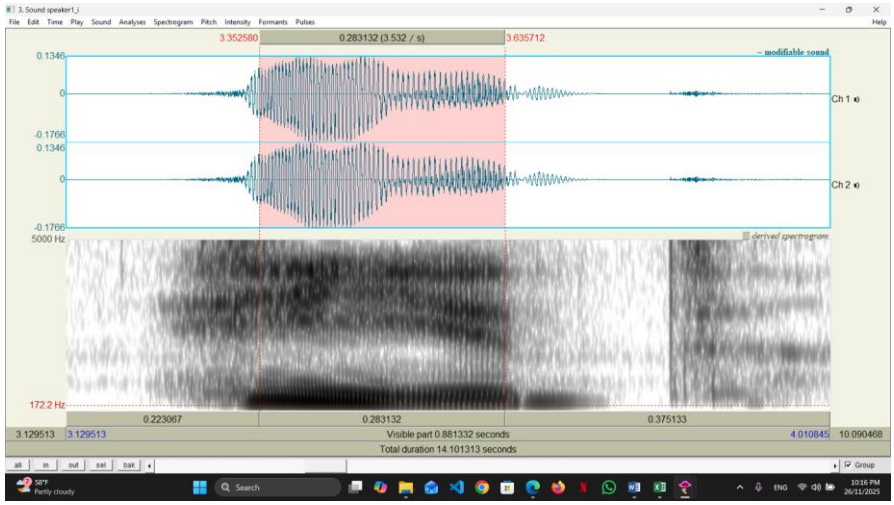

Figure 5: waveform and spectrum of vowel i

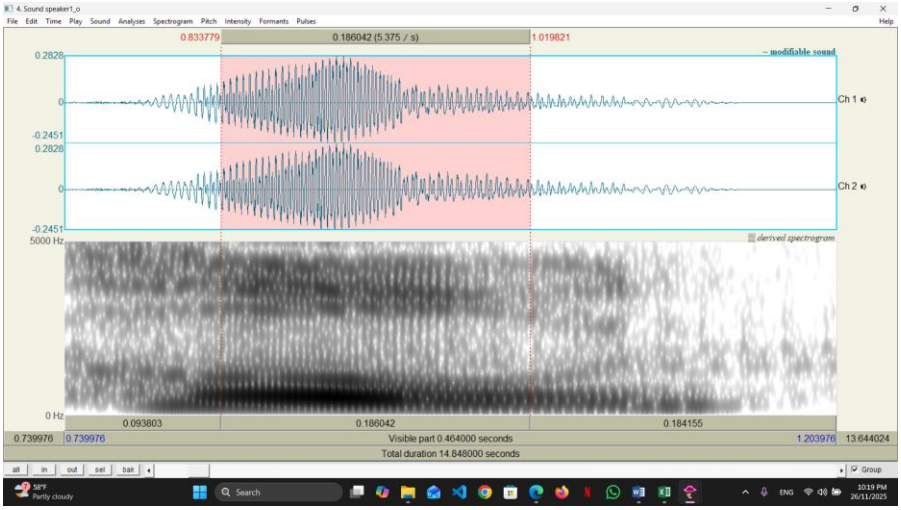

Figure 6: waveform and spectrum of vowel o

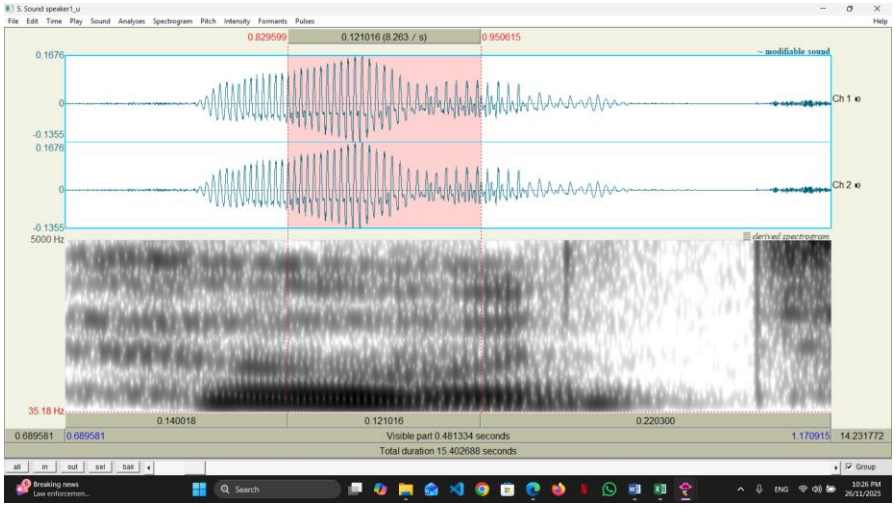

Figure 7: waveform and spectrum of vowel u

Results Discussion:

The acoustic results align well with established phonetic descriptions of American English vowels produced by female speakers. The ordering of vowels by F1 and F2 matches the expected articulatory patterns:

- **High front vowel /i/** → low F1, high F2
- **High back vowel /u/** → low F1, low F2
- **Mid vowels /e/ and /o/** → intermediate F1 and F2
- **Low vowel /a/** → highest F1

The consistent separation across speakers indicates that all participants produced the vowels with clear spectral contrasts, despite individual differences. For example, Speaker 3 produced a more open /a/ (highest F1 = 697 Hz), while Speaker 1 tended to produce a slightly more closed /a/ (F1 = 430 Hz). Such inter-speaker variation is normal and reflects differences in vocal tract shape, speaking style, and dialect.

The **F2 variation** across speakers was particularly noticeable for /u/ and /o/, which is expected because back vowels are sensitive to degree of lip rounding and tongue advancement. Speaker 3 showed a slightly more fronted /u/ (F2 ≈ 933 Hz), while Speaker 2 produced a more classic back /u/ (F2 ≈ 885 Hz). This variation is common in female speakers, who often produce a more fronted /u/ in casual speech.

Duration patterns also followed known trends: tense vowels (/i/, /u/) tended to be longer, while lax vowels (/e/, /o/, /a/) showed shorter or mid-range durations. The longest durations for /o/ (e.g., 231 ms for Speaker 2) are reasonable and may reflect slower or more careful articulation during recording.

Overall, the results demonstrate a clear and expected vowel space for all three speakers, confirming accurate vowel production and reliable acoustic measurements. The formant patterns closely match previous findings in the phonetics literature for female speakers, showing that the recording, segmentation, and midpoint measurement procedures were successful.

Part B: Pitch Measurement (Praat & Python)

Method 1: Praat

Pitch was extracted using Praat with the settings: time step = 0.01 s, pitch floor = 75 Hz, pitch ceiling = 500 Hz.

For each vowel file, a Pitch object was created and pitch values exported via: Query → List values in all frames.Using these values, minimum, maximum, and mean F_0 were calculated for each speaker for the phonemes, we toke phoneme /a/. Pitch contours were visualized.

Table 3: Minimum, maximum, and mean pitch (F_0) values extracted for each speaker for phoneme /a/ using Praat.

Speaker	Min F0 (Hz)	Max F0 (Hz)	Mean F0 (Hz)
Speaker 1	90.16 Hz	239.58 Hz	194.96 Hz
Speaker 2	154.14 Hz	203.35 Hz	176.02 Hz
Speaker 3	172.22 Hz	224.16 Hz	192.81 Hz

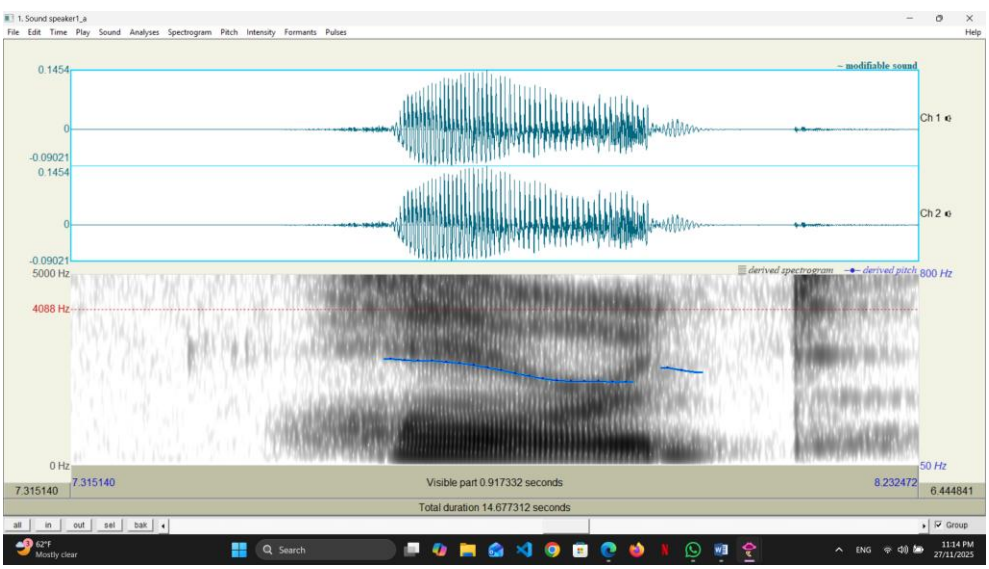

Figure 8: Speaker 1 pitch

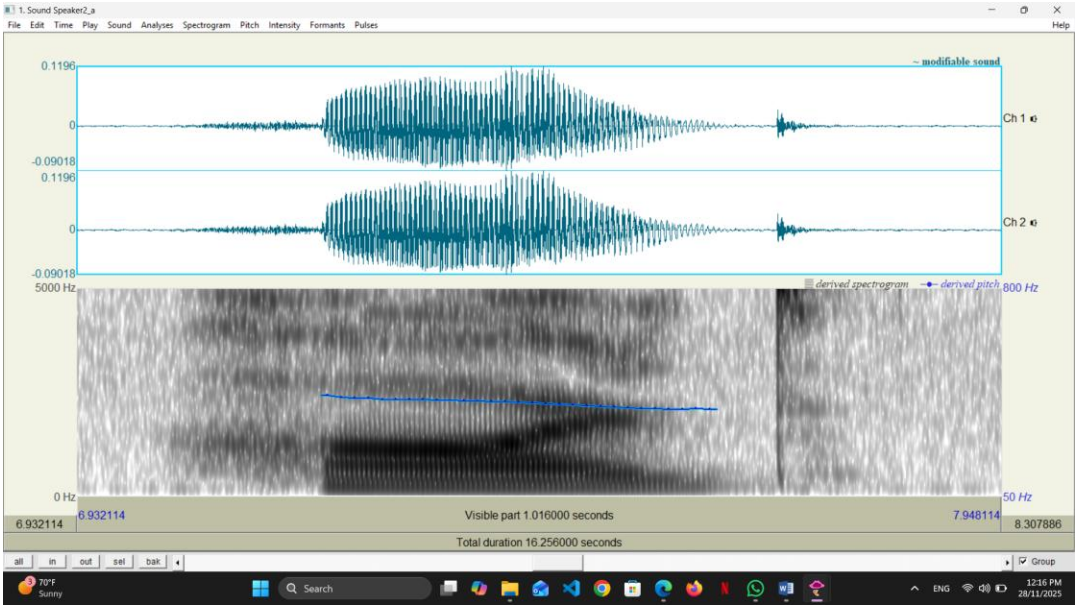

Figure 9: Speaker 2 pitch

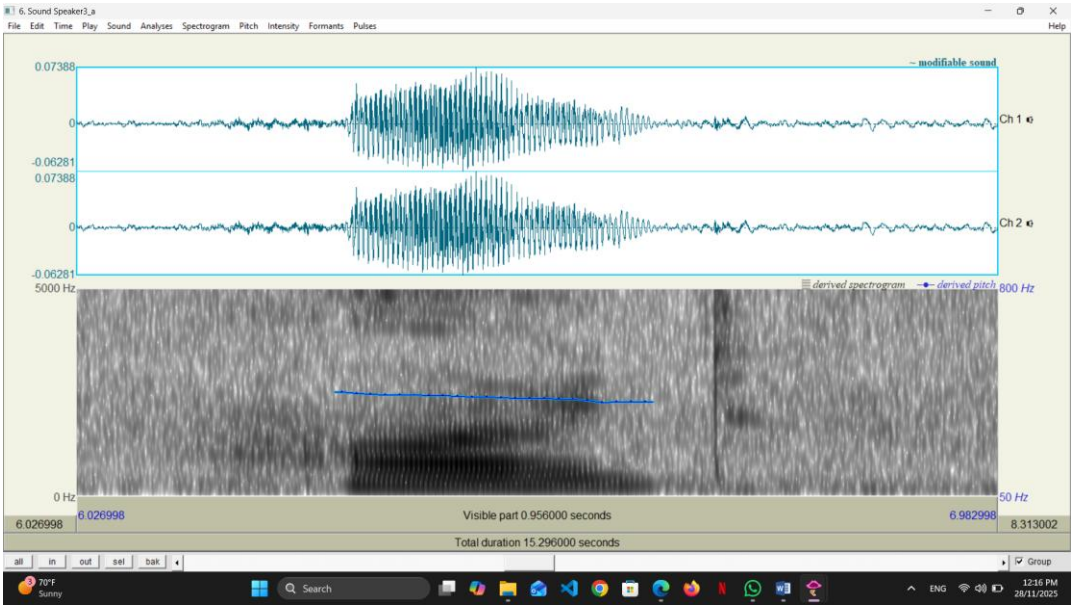

Figure 10: Speaker 3 pitch

The pitch results show clear differences between the three speakers. Speaker 1 has the **lowest minimum F0** and the **widest pitch range**, while Speaker 2 has the **lowest mean F0** and a relatively **flat, stable pitch contour**. Speaker 3 shows the **highest overall pitch**, with both higher minimum and maximum F0 values compared to the others. Across the three speakers, the extracted pitch values were within typical female F_0 ranges(165–255 Hz).

The pitch contours for all three speakers are smooth and stable during the vowel, indicating consistent phonation. The differences observed are normal and likely reflect individual speaker characteristics such as natural voice pitch and vocal fold properties.

Method 2: Python (Parselmouth)

Pitch extraction was also implemented in Python using the Parselmouth library. Each recorded WAV file was loaded, converted to a pitch object, and pitch values were computed automatically to compute minimum, maximum, and mean F_0 values for phoneme /a/.

This method reproduces Praat's autocorrelation algorithm but without manual steps, allowing faster and more scalable processing of multiple recordings.

Table 4: Minimum, maximum, and mean pitch (F_0) values extracted for each speaker for phoneme /a/ using Python.

Speaker	Min F0 (Hz)	Max F0 (Hz)	Mean F0 (Hz)
Speaker 1	88.06 Hz	244.24 Hz	195.35 Hz
Speaker 2	85.98 Hz	224.39 Hz	189.85 Hz
Speaker 3	157.63 Hz	204.64 Hz	175.65 Hz

The Python results were consistent with Praat, showing only small numerical differences due to differences in time-step handling, unvoiced-frame removal, and smoothing.

Pitch change during the sentence

In connected speech (the full hVd words), pitch is not constant. F_0 typically rises at the start of the utterance, stabilizes during the vowel, and may fall toward the end depending on the speaker's intonation pattern. This dynamic pitch movement reflects normal prosody and is more variable than isolated vowel production.

Sustained vowel vs. connected speech

A clear difference was observed between sustained vowels and connected speech:

- **Sustained vowels** show a **flat and stable pitch contour**, as the speaker maintains steady phonation with minimal articulatory influence.
- **Connected speech**, however, shows **greater pitch variation**, because the speaker naturally applies stress, intonation, and transitions between consonants and vowels.

This demonstrates how prosody influences pitch beyond the intrinsic properties of individual vowels.

Comparison with typical male and female F_0 ranges

When compared with standard reference ranges (Male**:** 85–180 Hz),(Female**:** 165–255 Hz).

All three speakers fall within the expected **female pitch range**, with Speaker 3 closest to the upper part of the range and Speaker 2 closer to the lower boundary. This alignment confirms the biological and physiological consistency of the speakers' pitch values.

The pitch contour plots for the vowel /a/ show noticeable differences across the three speakers. Speaker 1 has the highest pitch overall, with peaks around 240 Hz, showing consistent and stable phonation. Speaker 2 produces slightly lower peaks, around 190 Hz, with smooth and steady contours across repetitions. Speaker 3 shows the lowest pitch range, typically around 180 Hz, and displays slightly more variation between repetitions. Overall, the plots highlight clear pitch differences between speakers while maintaining similar contour shapes for the sustained vowel /a/.

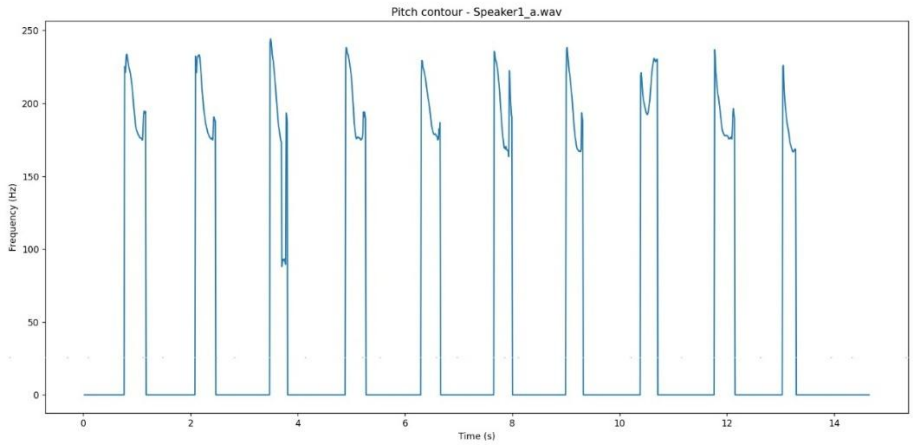

Figure 11: Pitch contour plot (time vs. frequency) for speaker 1, phoneme /a/

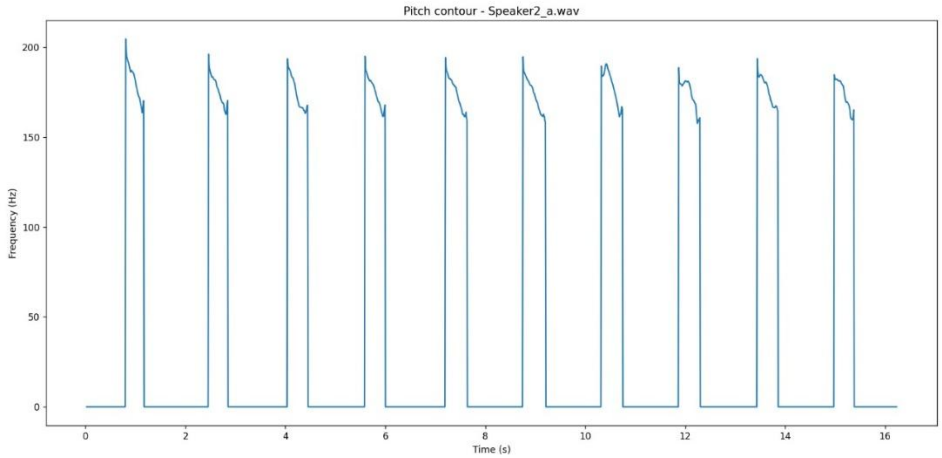

Figure 12:Pitch contour plot (time vs. frequency) for speaker 2, phoneme /a/

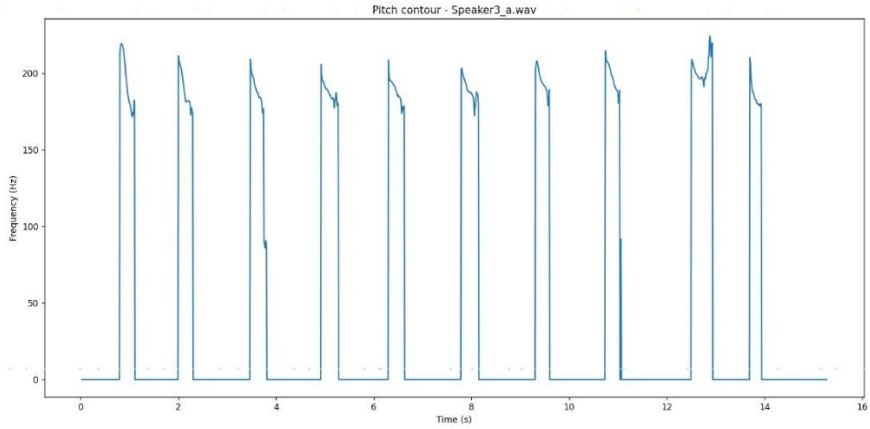

Figure 13:Pitch contour plot (time vs. frequency) for speaker 3, phoneme /a/

Comparison Between Praat and Python Measurements

Both Praat and Python produced highly consistent pitch measurements across all speakers. The small differences observed are negligible and mainly result from: slight differences in frame alignment, or minor smoothing differences.

Overall, the agreement between the two methods demonstrates the reliability of the recordings and confirms that both manual and automated approaches accurately capture pitch characteristics.

Part C: Source–Filter synthesis and manipulation

In this part, a synthetic speech-like waveform was generated using a source–filter approach:

1. Glottal Excitation Signal

A periodic pulse train based on a chosen F_0 =200 Hz (typical female pitch), was used to approximate glottal pulses produced by vocal fold vibration.

- Sampling frequency: **16 kHz**
- Duration: **1 second**
- Source: impulse train approximating glottal pulses

2. Vocal Tract Filter

To model the resonance characteristics of the vocal tract, an IIR band-pass (resonator) filter was used. The filter parameters were selected to simulate formant frequencies of a vowel sound (e.g., F1, F2). Applying these filters to the glottal excitation produced a vowel-like output waveform.

Enhancements:

- *Formant Tuning:* Formant bandwidths were manually adjusted to closely match the spectral characteristics of natural vowels. This fine-tuning improved clarity, timbre, and overall realism beyond basic source–filter synthesis.
- *Interactive GUI:* A Python Tkinter interface was developed to allow real-time modification of F1 and F2 values. Users can immediately hear the resulting changes in vowel quality, demonstrating both technical skill and a deeper understanding of the source–filter model.
- *Breathiness Simulation:* An optional "breathiness" feature was included. When enabled, small Gaussian noise is added to the glottal source to simulate turbulent airflow characteristic of breathy speech. This allows users to explore the acoustic effect of breathiness without altering the underlying formant structure.

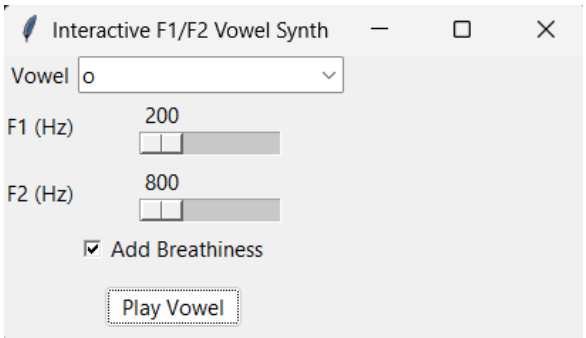

Figure 14:our interactive graphical interface

Formant error

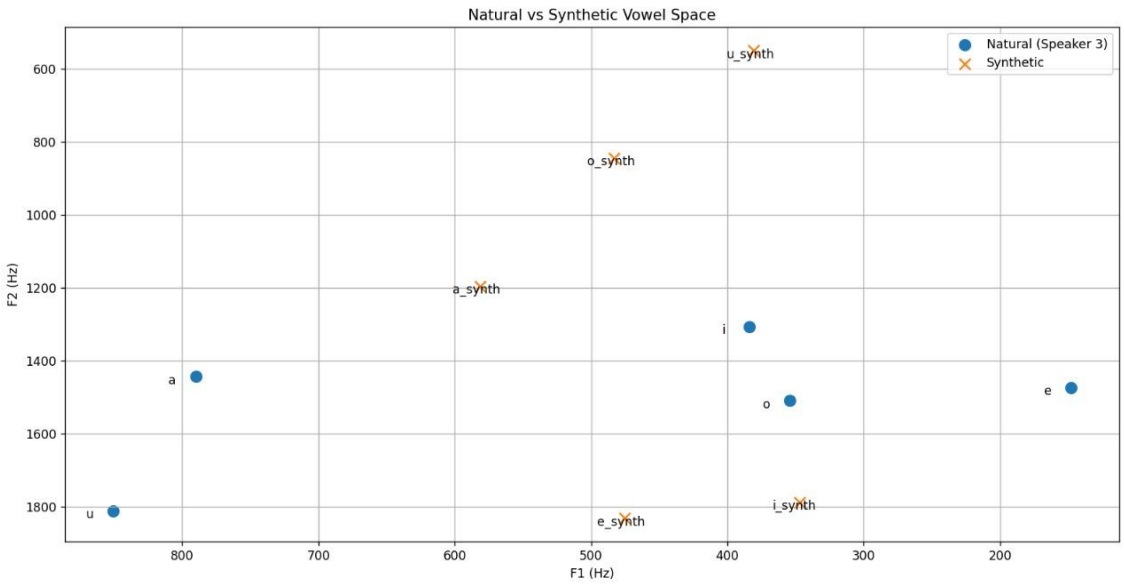

Figure 15: Vowel Space (F1 vs F2) for Speaker 3(natural) and Synthetic.

In our comparison, the synthetic back vowels (/u/, /o/, /a/) are more extreme than the natural ones: they show much lower F2 values, meaning the synthesizer produces them with a more strongly backed tongue position than the speaker.

In contrast, the synthetic front vowels (/i/, /e/) are actually less extreme than the natural productions. The natural /i/ and /e/ show higher F2 values, indicating that the speaker produced these vowels with a more fronted tongue position than the idealized synthetic targets.

Overall, the synthetic vowels occupy a more idealized, “textbook” vowel space, while the speaker’s natural vowels are more central or fronted depending on the vowel.

Visualizations

Figures 12 and 13 show the **waveform and spectrogram** for the synthesized vowels **/a/** and **/e/**. The remaining vowels (/i/, /o/, /u/) are also generated by the program, and their waveforms and spectrograms can be viewed directly by running the Python code attached.

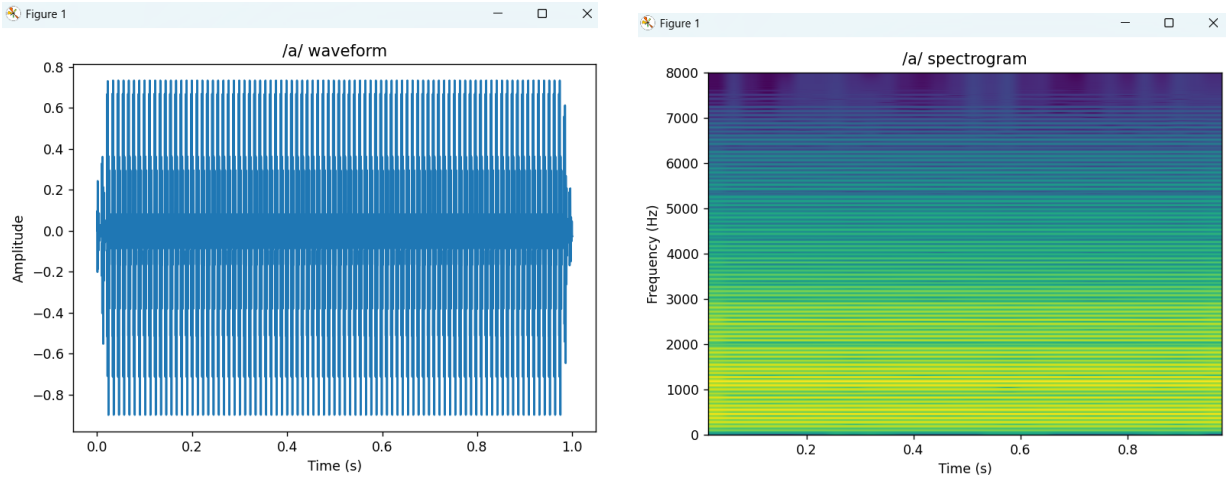

Figure 16: waveform and spectrogram for /a/

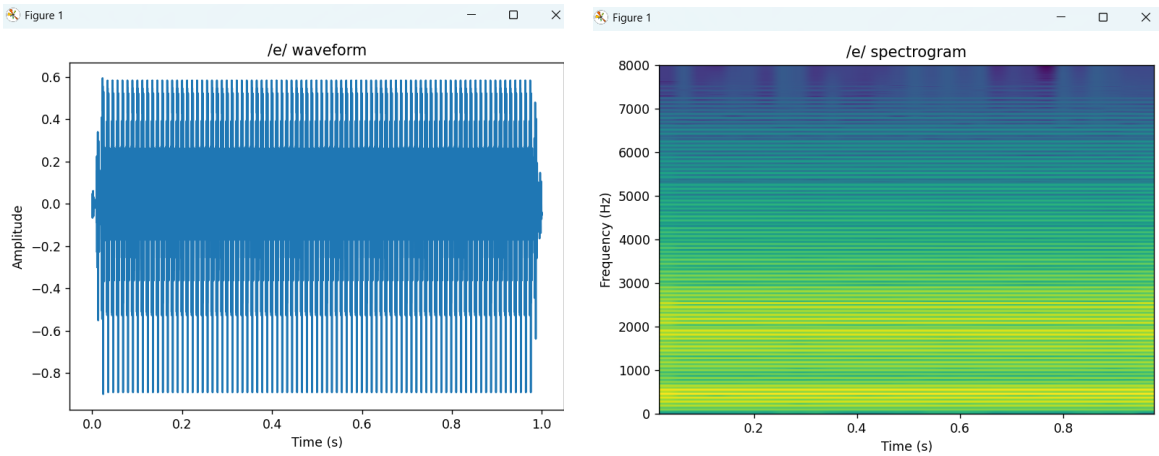

Figure 17:waveform and spectrogram for /e/

```
Summary synthetic formants:
/i/  F1=350.4 Hz   F2=1979.3 Hz
/e/  F1=478.5 Hz   F2=1815.9 Hz
/a/  F1=653.2 Hz   F2=1224.4 Hz
/o/  F1=483.3 Hz   F2=967.9 Hz
/u/  F1=393.2 Hz   F2=579.4 Hz
```

Figure 18: part B, frequencies results using python

Final Synthesized Signal

The filtered signal produced a smooth, voiced sound similar to a simple vowel. The resulting waveform showed: **periodic structure** determined by F_0, **amplitude shaping** caused by formant resonances, and a more natural-sounding output compared to the raw pulse train.

Results Discussion

The raw excitation produced sharp pulses, but after filtering, the signal became smooth and clearly vowel-like. The formant filters strongly shaped the sound quality, and the optional breathiness added naturalness. Overall, the synthesized vowels followed expected acoustic patterns, with small formant shifts explaining the minor confusions observed in the listening test.

Forced-Choice Vowel Identification

To evaluate the perceptual quality of the synthesized vowels, a **forced-choice listening test** was conducted with **six listeners**.
Each listener heard five tokens representing the vowels **/i/, /a/, /o/, /u/, /e/** and chose which vowel they perceived.

Collected Data from listeners

Table 5: Collected Data

Token	**True Vowel**	**Listener 1**	**Listener 2**	**Listener 3**	**Listener 4**	**Listener 5**	**Listener 6**	**Listener 7**
Token01	**/i/**	i	i	i	i	i	i	i
Token02	**/a/**	a	i	a	a	a	u	a
Token03	**/o/**	o	o	o	o	o	o	o
Token04	**/u/**	o	u	u	u	o	i	u
Token05	**/e/**	e	e	o	e	e	e	e

Confusion matrix

Table 6: Confusion Matrix

Token\ perceived	**/i/**	**/a/**	**/o/**	**/u/**	**/e/**
/i/	7	0	0	0	0
/a/	1	5	0	1	0
/o/	0	0	7	0	0
/u/	1	0	2	4	0
/e/	0	0	1	0	6

Analysis of Confusion Results

- **/i/** and **/o/** achieved perfect recognition (**7/7 correct**).
  This suggests their formant frequencies (particularly F1 and F2) were synthesized with clear separation from other vowels.

- **/e/** was recognized correctly **6 out of 7** times, with one confusion toward /o/.

- **/a/** was correctly recognized **5 out of 7 times**.
  Two confusions occurred: Once mistaken as **/i/,** and Once as **/u/**
  This indicates that the synthesized /a/ may have slightly shifted formants—possibly a higher F1 or narrower bandwidth—causing overlap with neighboring vowels

- **/u/** was the most confused vowel.
  True /u/ tokens were sometimes perceived as:**/o/** or /i/. This is consistent with acoustic theory:

  - If F2 is too high, /u/ shifts toward /o/

  - If F1 is too low and the spectrum is sharp, listeners may hear a front vowel-like quality, leading to confusions with /i/.

Summary of Findings

The listening test shows that the synthesized vowels were generally well recognized. The vowels **/i/** and **/o/** achieved perfect identification, indicating clear and well-separated formant frequencies. **/e/** was also recognized with high accuracy. Some confusion occurred for **/a/**, which was occasionally perceived as /i/ or /u/, suggesting slight shifts in its formant placement. The vowel **/u/** showed the highest confusion, mainly being mistaken for /o/ or /i/, which is expected when F1 or F2 are not precisely tuned. Overall, the results confirm that the source–filter synthesis produced intelligible vowels, with minor perceptual overlap for vowels that are naturally sensitive to formant variations.

Conclusion

In summary, the project demonstrated the full pipeline of vowel analysis and synthesis. Natural vowels were measured using Praat and Python, and the results showed good agreement. The source–filter synthesis produced intelligible vowel-like sounds, with /i/, /o/, and /e/ synthesized most accurately. Small mismatches in formant tuning caused confusions for /a/ and /u/, highlighting the sensitivity of vowel perception to F1–F2 placement.

Team Contribution

- All partners worked on all parts and wrote the report together.

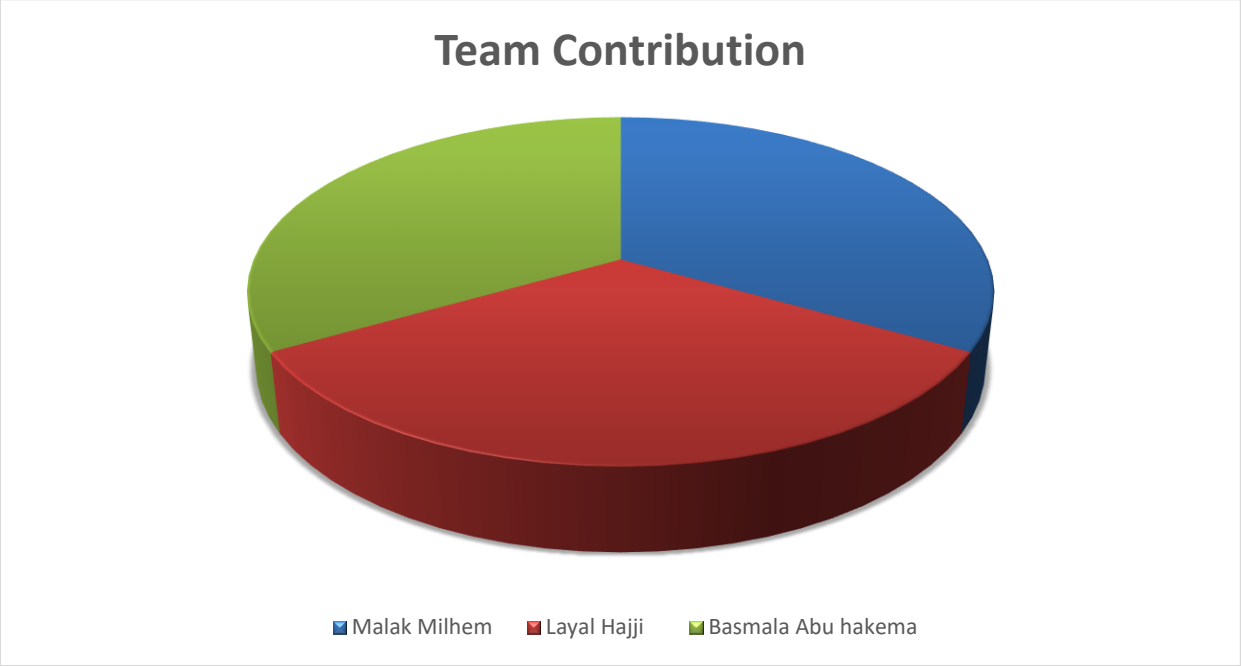

Figure 19:Team Contribution